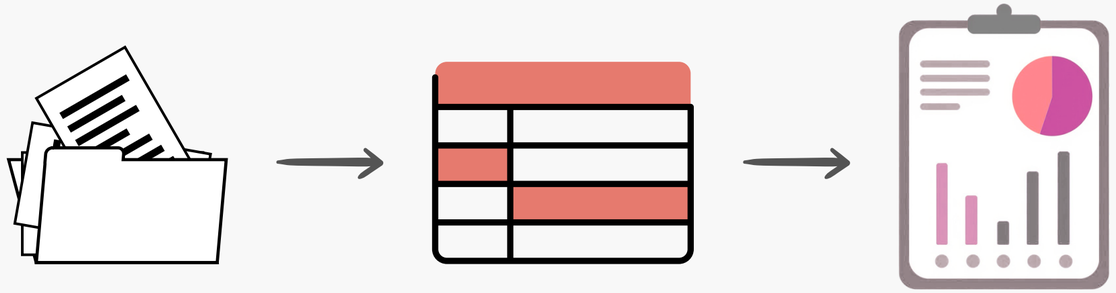
GitHub
Tableau

ROAD TRAFFIC ACCIDENT SEVERITY ANALYSIS

PUBLIC SAFETY & ROAD TRAFFIC MANAGEMENT



Data Preparation (**PYTHON**) → Data Analysis (**EDA**) → Insights Visualization (**TABLEAU**)

TEAM MEMBERS

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CONTEXT & PROBLEM STATEMENT

CONTEXT

> Ethiopia has high road fatality rates due to rapid urbanisation + weak enforcement

Decision-makers: Traffic police & safety authorities

PROBLEM STATEMENT

What factors make accidents **serious** or **fatal**?

OBJECTIVE

Identify key risk factors → help allocate enforcement resources effectively

12,295
Accident Records analyzed

1.29%
Fatality Rate

17:00
Peak Accident Time

DATA ENGINEERING

DATA SOURCE

Ethiopian RTA dataset

ORIGINAL VOLUME		FINAL DATASET
12,316	→	**12,295**
Total Rows		Rows

DATA QUALITY

99.83% retained | 0 nulls | 0 duplicates

01 Data Collection

Collected 12,316 accident records from Ethiopian RTA dataset (Kaggle)

02 Cleaning & EDA

Handled missing values, removed inconsistencies, and explored key patterns in data

03 Analysis

Identified major risk factors and computed key KPIs driving accident severity

04 Visualization

Built interactive Tableau dashboard for decision-making and insights

KPI & METRICS FRAMEWORK

TOTAL ACCIDENTS

12,295

TOTAL CASUALTIES

19,035

FATAL RATE

1.29%

MOST AT-RISK GROUP

18-30 (MALE)

WEEKEND RISK

2.45×

WORST TIME PERIOD

17:00 (NIGHT)

How are they computed?

- Derived from cleaned dataset (12,295 records)
- Calculated using Python (Pandas)
- Stored in: 05_final_load_prep.ipynb → kpi_summary.csv

KEY INSIGHTS (EDA)

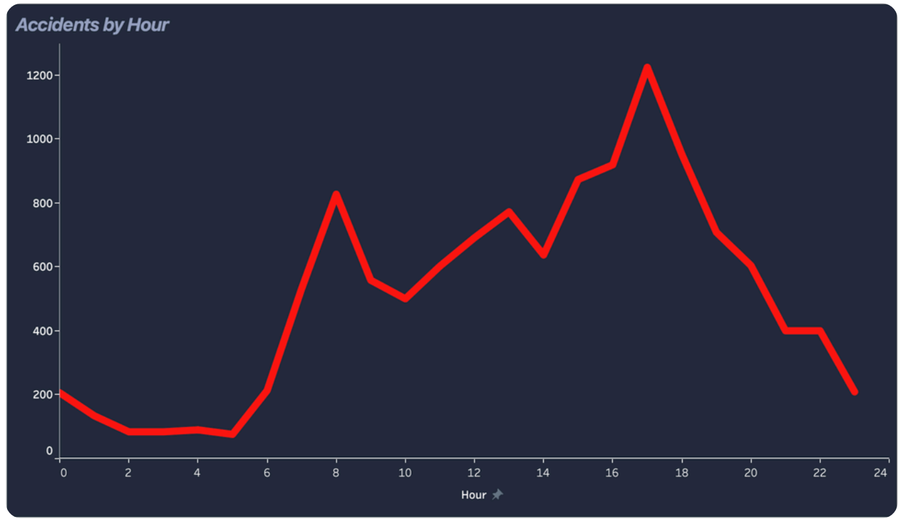
Weekend accidents are **2.45x** more fatal

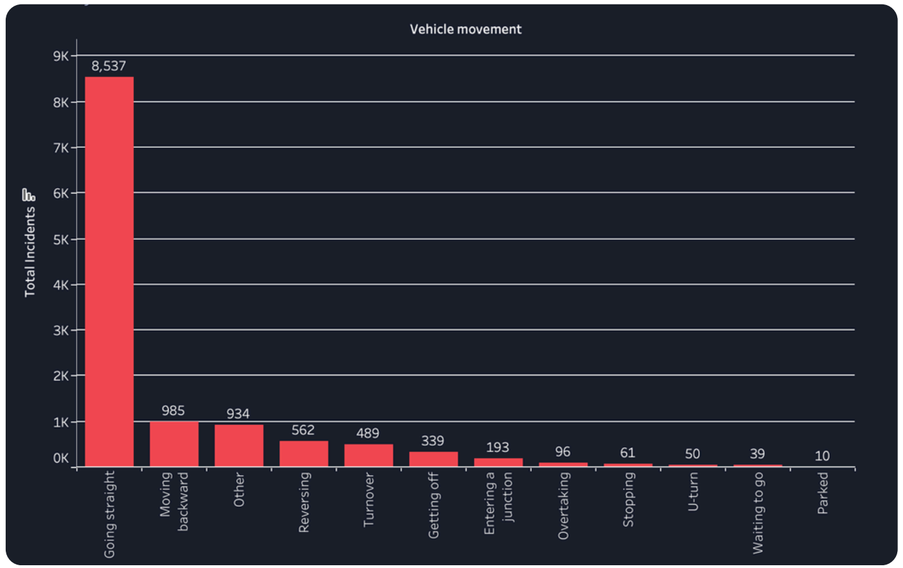
84% accidents happen in normal conditions → behaviour issue

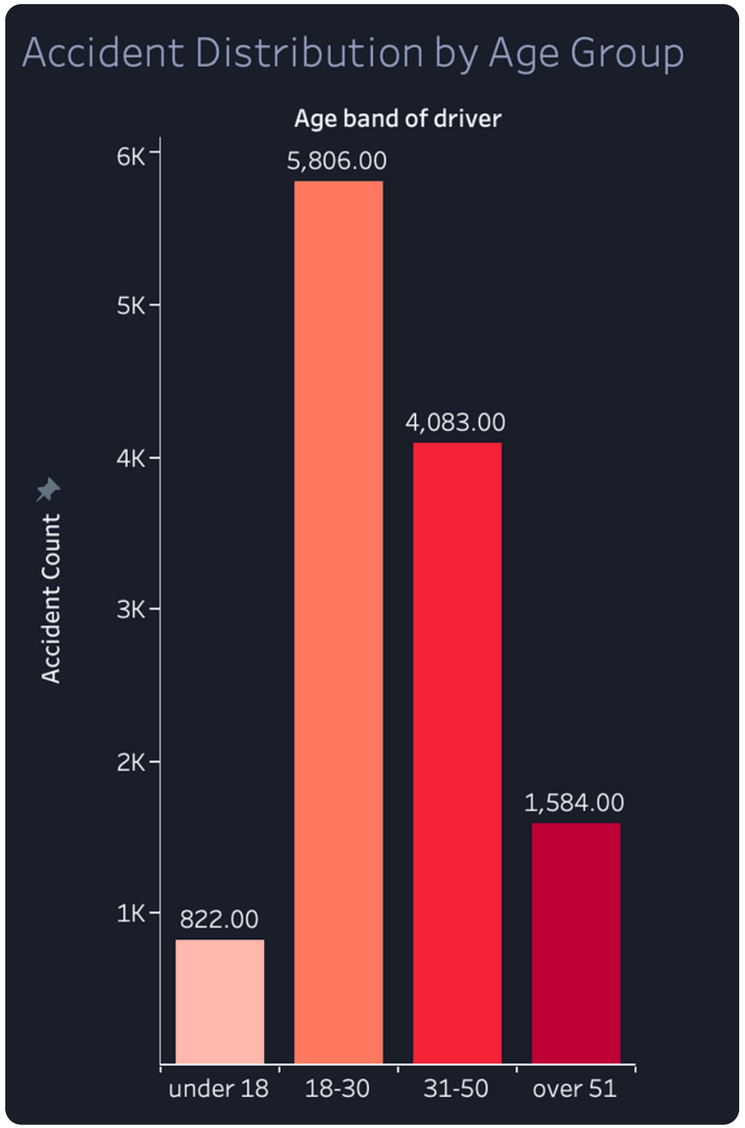
17:00 is peak accident hour

18–30 age group → **47%** of drivers involved

No distancing causes **18.6%** accidents







ADVANCED ANALYSIS

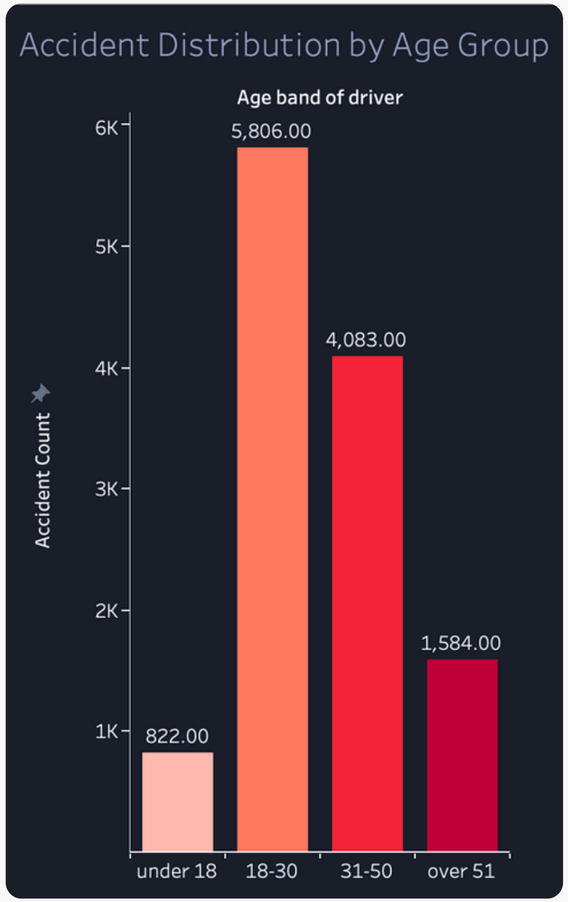
ANALYTICAL DEPTH (WHAT WE DID)

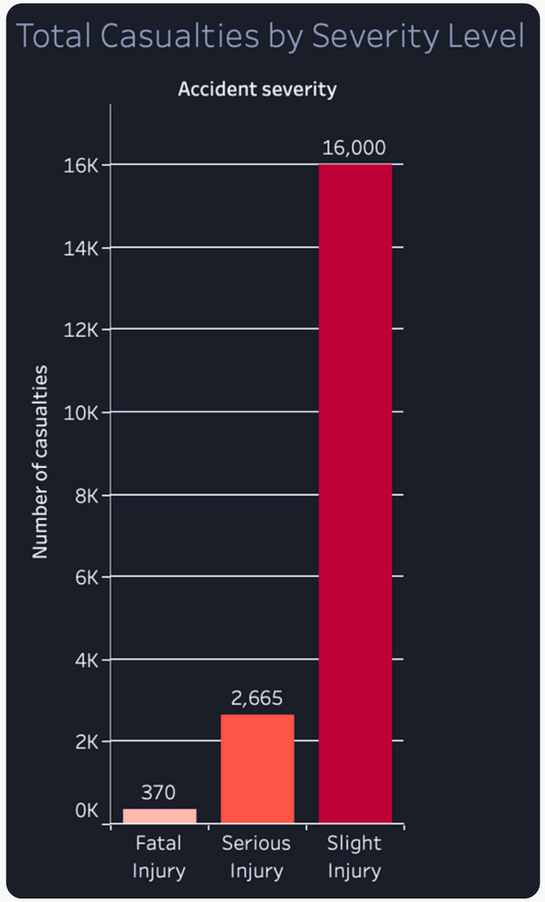
- Segmentation Analysis → Driver age, experience, and vehicle type
- Root Cause Analysis → Identified key behavioral factors (distancing, lane violations)
- Comparative Analysis → Weekend vs weekday, day vs night risk patterns

NEW UNDERSTANDING (BEYOND EDA)

- Mid-experience drivers (**2–5 yrs**) show highest fatal risk → **overconfidence** effect
- **Night-time** accidents are most severe per incident, not just frequent
- Majority accidents occur in normal conditions → **behaviour** is primary cause, not environment
- **Heavy** vehicles significantly increase severity risk

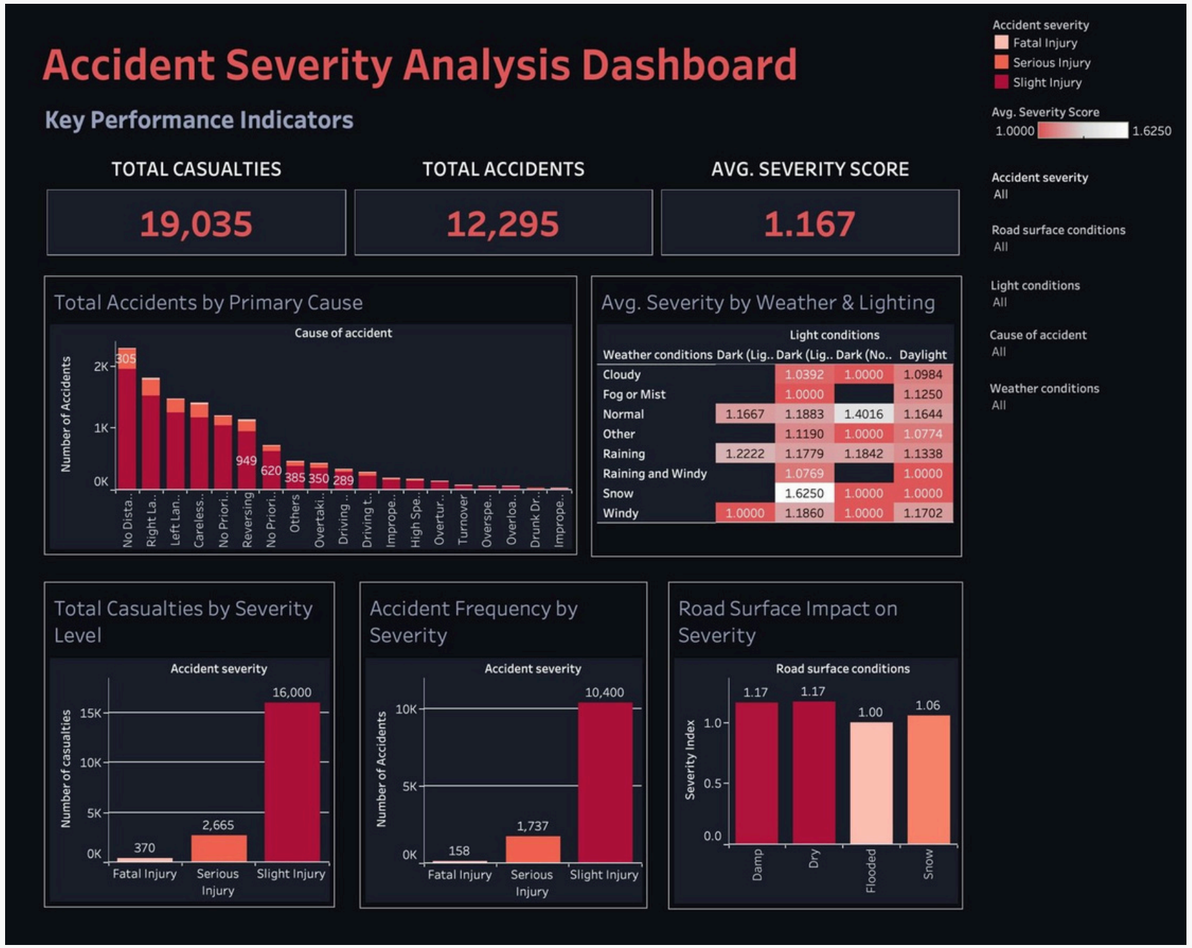
Python (Pandas, NumPy)



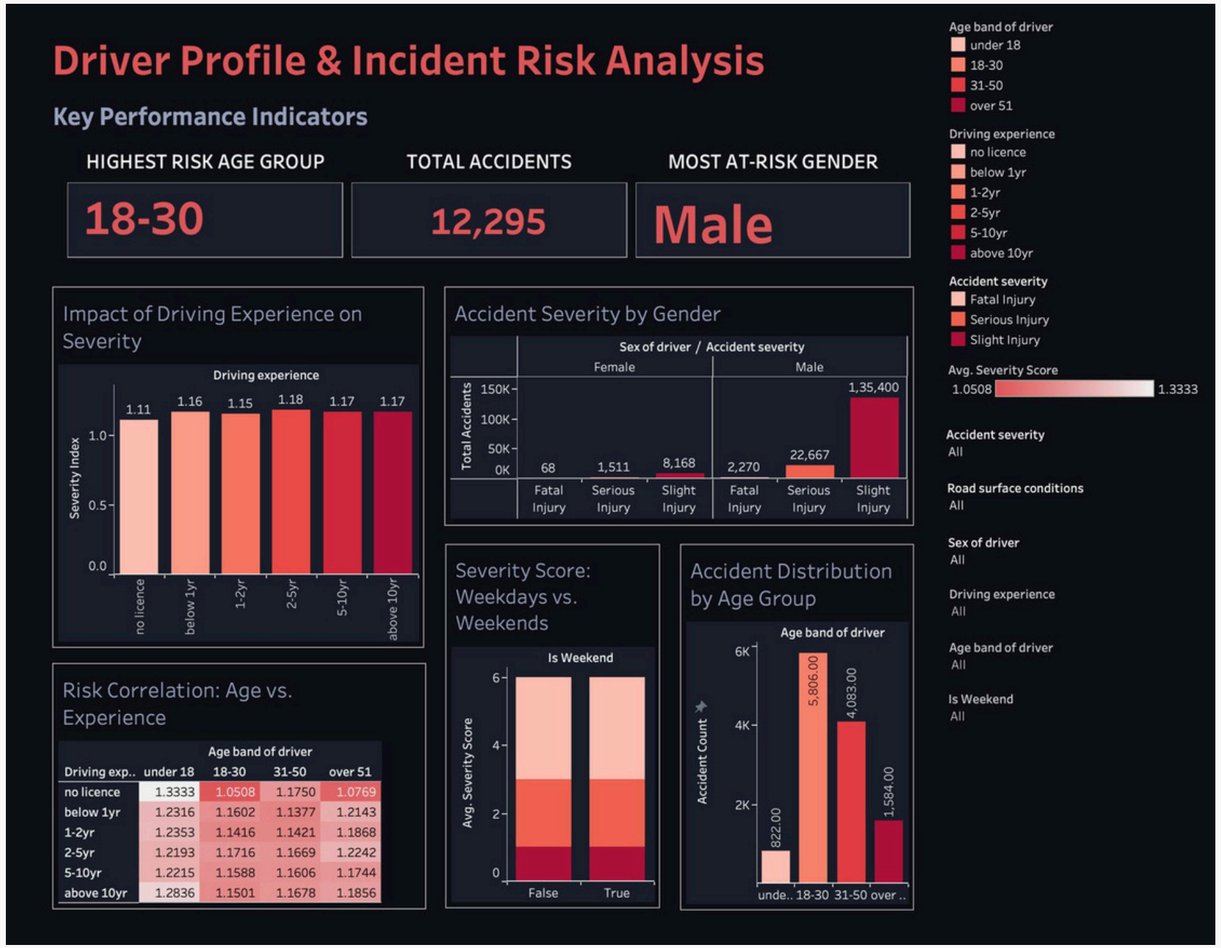


DASHBOARD WALKTHROUGH

Link



DASHBOARD 1



DASHBOARD 2

DASHBOARD WALKTHROUGH

DASHBOARD 3

DASHBOARD 4

RECOMMENDATIONS

01.

Weekend Patrol Increase

→ Focus Fri evening + weekends
→ Reduce fatal accidents

02.

Young Driver Training

→ Mandatory hazard training
→ Reduce crashes in 18–30

03.

Distancing Enforcement Cameras

→ Target top junctions
→ Reduce violations

04.

Heavy Vehicle Control

→ Monitor trucks/lorries
→ Reduce severe accidents

IMPACT & VALUE

Reduced Fatal Accidents

Targeted weekend and night enforcement can reduce fatality rates by **~20%**

Efficient Resource Allocation

Focus on **peak hours** (**16:00–22:00**) improves patrol effectiveness

Behaviour-Based Intervention

Addressing key causes (distancing, lane violations) can impact **~45%** of accidents

Data-Driven Decision Making

Dashboard enables faster, more accurate policy and enforcement decisions

Impact Snapshot

- **20%** ↓ fatal accidents
- **2.45×** risk addressed
- **45%** accidents targetable
- **0** extra cost (some actions)

LIMITATIONS & NEXT STEPS

⚠️ LIMITATIONS

Class imbalance → only **1.29%** fatal cases (limited statistical depth)

No geolocation data → cannot identify exact accident hotspots

No time trend (year-wise) → cannot measure changes over time

Reporting bias → minor accidents may be underreported

🚀 FUTURE ROADMAP

Predictive Model → accident severity prediction (ML)

Geospatial Analysis → hotspot detection using GPS data

Real-time Dashboard → live accident monitoring system

Extended Dataset → include multi-year and location-based data

TEAM & CONTRIBUTION

Team Member	Role	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Suhaani Garg	Project Lead	Y	Y			Y	Y	
Krishna Dave	Strategy Lead				Y	Y		
Sathvik	ETL Lead	Y	Y			Y		
Hardik Maheshwari	Analysis Lead			Y	Y			
Anushka Tyagi	PPT & Quality Lead					Y		Y
Keshav	Data Lead			Y			Y	
Arya Pandey	Visualization Lead				Y	Y		